COMPUTER SCIENCE DOCUMENTATION PARTS

A structured guide for software, system, and capstone project documentation

Purpose

This guide presents the common sections used in documenting a Computer Science project. The exact structure may be adjusted according to institutional or adviser requirements.

1. Preliminary Pages

- Title Page
- Approval Sheet
- Acknowledgment
- Abstract
- Table of Contents
- List of Tables
- List of Figures

2. Chapter 1 - Introduction

- Background of the Study
- Statement of the Problem
- Objectives of the Project: General Objective and Specific Objectives
- Scope and Limitations
- Significance of the Project
- Definition of Terms

3. Chapter 2 - Review of Related Literature and Systems

- Related Literature
- Related Studies
- Related Systems or Applications
- Comparison of Existing Systems
- Synthesis
- Research or System Gap
- Conceptual Framework

4. Chapter 3 - Methodology and System Design

- Software Development Methodology
- Requirements-Gathering Methods
- System Requirements: Functional and Nonfunctional Requirements
- System Architecture
- Use-Case Diagram
- Data-Flow Diagram
- Entity-Relationship Diagram
- Database Design
- Class Diagram
- Activity or Sequence Diagram
- User-Interface Design
- Hardware and Software Requirements
- Development Tools and Technologies

5. Chapter 4 - System Development, Results, and Discussion

- System Implementation
- Description of System Modules
- User-Interface Presentation
- Database Implementation
- Algorithms and Important Code
- System Testing
- Test Cases and Results
- Performance Evaluation
- Security and Data-Privacy Measures
- Discussion of Findings

6. Chapter 5 - Summary, Conclusions, and Recommendations

- Summary of the Project
- Conclusions
- Recommendations
- Proposed Future Enhancements

7. References

- Books
- Journal Articles
- Websites
- Technical Documentation
- Other sources using the required citation style

8. Appendices

- Project Timeline or Gantt Chart
- Budget
- Survey or Interview Questions
- Requirements Specification
- Test Cases
- Sample Source Code
- User Manual
- Installation Guide
- Screenshots
- Letters and Permissions
- Data-Privacy Consent Form
- Researchers' Profiles or Curriculum Vitae

Additional Technical Documents

- Software Requirements Specification (SRS)
- Software Design Document
- Database Documentation
- API Documentation
- Source-Code Documentation
- Testing and Quality-Assurance Document
- Deployment and Installation Guide
- System Administrator Manual
- User Manual
- Maintenance and Troubleshooting Guide
- Version History and Change Log